Data Scientist

Leveraging Machine Learning and Analytics to Drive Data-Informed Decisions

Machine Learning | Data Analytics | Python | SQL | Predictive Modeling

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About Me

digging into messy data and turning it into real, useful insights with machine learning and stats. I've got solid skills in both hands-on coding and sharp problem-solving, and I'm all about building reliable data pipelines and predictive models that actually fix real business headaches. Right now, I'm hunting for full-time Data Scientist roles starting Fall 2026.

Education

- **Lingnan University** | Master of Science in Data Science | Expected Graduation: August 2026

My Strengths / Focus Areas

- **End-to-End Data Modeling:** I take fuzzy business problems and turn them into full machine learning projects – from cleaning data and picking features to getting the model live and running.
- **Analytical Problem Solving:** I dig into model results, question every assumption, and check everything twice to keep the data solid and the answers accurate.
- **Technical Independence:** I write, debug, and speed up all the code myself for data work and analysis, so it scales nicely without needing extra help.

Skills

- **Programming:** Python, SQL
- **Data Science & ML:** Machine Learning, Predictive Modeling, Statistical Analysis, Natural Language Processing (NLP), Deep Learning
- **Business Skills:** Problem Framing, A/B Testing & Experimentation, Data Visualization & Storytelling, Metrics Definition
- **Tools & Frameworks:** Pandas, NumPy, Scikit-learn, TensorFlow / PyTorch, Tableau / Power BI, Git, GitHub

Projects (Portfolio)

End-to-End Gait Stability and Fatigue Detection using Wearable IMU Sensors

Time: [Jan 2026 – Apr 2026]

- **Problem:** Identifying subtle human gait abnormalities is critical for occupational safety and healthcare. The objective was to build a robust predictive system capable of classifying an individual's real-time walking state into three categories (Normal, Distraction, Fatigue) using continuous wearable sensor data.
- **Data:** Utilized the "duo" dataset consisting of 6-axis IMU time-series data (3D Accelerometer and 3D Gyroscope) collected from a lower back/sacrum sensor, resampled to 128Hz.
- **Approach:**
  - **Signal Processing:** Implemented robust data pipelines including linear interpolation for missing values, median filtering, and Butterworth low-pass filtering.
  - **Feature Engineering:** Extracted complex spatial-temporal, kinematic, and frequency-domain features (harmonic ratio). Innovated a "relative feature" algorithm using the first 30 seconds of normal walking as a personalized baseline to prevent data leakage.
  - **Machine Learning:** Developed Generalized and Personalized XGBoost classifiers. Optimized hyperparameters using Optuna and employed StratifiedGroupKFold for robust validation. Conducted SHAP value analysis for interpretability.

- **Outcome/Impact:** The personalized models improved the average accuracy to **0.966**.
- **Contribution:** Sole developer. Independently architected and implemented the entire pipeline.
- **Links:** https://github.com/hecaizhis-dev/gait_stability_preview

Credit Card Fraud Detection using Advanced Feature Engineering

Time: [Feb 2026 – Mar 2026]

- **Problem:** The objective was to build a highly sensitive predictive model to identify fraudulent transactions in real-time, effectively balancing Recall and Precision in a highly imbalanced environment.
- **Data:** The Kaggle Credit Card Fraud Detection dataset (284,807 transactions, 0.172% fraud rate).
- **Approach:**
  - **EDA & Feature Engineering:** Conducted deep visual analysis (Seaborn). Extracted temporal signals (midnight/nightfall indicators) and rolling transaction frequencies. Created interaction features combining high-correlation PCA variables.
  - **Machine Learning Pipeline:** Designed an imblearn pipeline integrating StandardScaler, SMOTE, and an XGBoost classifier configured with scale_pos_weight. Automated hyperparameter tuning via Optuna.
- **Outcome/Impact:** The model achieved a robust ROC-AUC of **0.98** and an optimized F1-score of **0.82**.
- **Contribution:** Sole developer. Conducted the end-to-end process from EDA to model evaluation.
- **Links:** https://github.com/hecaizhis-dev/CreditCardFraudDetection-ML

March Machine Learning Mania 2026 (NCAA Basketball Prediction)

Time: [Mar 2026]

- **Problem:** Predicting the probabilities of team matchups in the NCAA Men's Basketball Tournament while strictly preventing future data leakage.

- **Data:** Official Kaggle dataset including historical regular season detailed results and tournament seeds.
- **Approach:**
  - **Point-in-Time (PIT) Architecture:** Engineered a rigorous dual-state dictionary to iteratively track cumulative team statistics match-by-match, eliminating data leakage.
  - **Feature Engineering:** Constructed derived metrics aligned with basketball analytics (eFG%, TOV%, ORB%, FTR%) and developed a dynamic Elo rating system.
  - **Modeling:** Deployed a Logistic Regression model via scikit-learn pipeline, evaluated using Brier Score Loss and ROC-AUC.
- **Outcome/Impact:** Established a solid, leakage-free predictive pipeline capable of generating submission-ready probabilities for the Kaggle format.
- **Contribution:** Sole developer. Designed the PIT data tracking logic and the dynamic Elo rating system.

Resume / CV

Please view or download my complete resume below:

📄 **Download My Resume (PDF)**

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